INCH-POUND

MIL-DTL-17587/8(SH)
28 June 2007

DETAIL SPECIFICATION SHEET

CIRCUIT BREAKERS, ACB, LOW VOLTAGE, ELECTRIC POWER, AIR, REMOVABLE CONSTRUCTION ACB-901R, ACB-2601R, ACB-2603R, ACB-2801R, ACB-4001R, AND ACB-5003R

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17587.

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REQUIREMENTS:

TABLE I. Ratings/features of circuit breakers.

Circuit breaker type	ACB-901R	ACB-2601R	ACB-2603R	ACB-2801R	ACB-4001R	ACB-5003R
Rated continuous current (amperes)	900	2600	2600	2800	4000	5000
Rated maximum voltage (DC volts) /1	355	355	355	355	355	800
Maximum interrupting rating current (amperes)	50,000	75,000	150,000	25,000	150,000	75,000
Rated short-time current /2 (amperes)	25,000	50,000	25,000	25,000	150,000	75,000
Rated short-time current duration (seconds)	0.5	0.5	0.5	0.5	0.5	0.5
Operating mechanism /3 - manual	STD	STD	STD	STD	STD	STD
Operating mechanism - electric	STD	STD	STD	STD	STD	STD
Maximum weight (lbs) - circuit breaker (removable component)	192	447	218	520	424	500
Maximum weight (lbs) - cradle (stationary component)	104	193	112	215	173	185
Number of poles	2	2	2	3 /3	2	2
Number of overcurrent coils or solid-state sensors	2	2	2	2	2	2
Attachments						
Electric closing mechanism	OPT	STD	STD	STD	STD	STD
Auxiliary switches (number of contacts)	6	8	8	8	12	12
Undervoltage lockout device	Optional for all circuit breaker types.					
Mechanical position indicator	Optional for all circuit breaker types.					
Secondary disconnecting devices	Required for all circuit types, unless otherwise specified.					
Cell switch	Optional for all circuit breaker types.					
Indicator light	Provisions for mounting required for all circuit breaker types; furnished only when specified.					

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TABLE I. Ratings/features of circuit breakers - Continued.

Circuit breaker type	ACB-901R	ACB-2601R	ACB-2603R	ACB-2801R	ACB-4001R	ACB-5003R
Hold-closed mechanism	OPT	OPT	OPT	OPT	OPT	OPT
Undervoltage release device	Optional for all circuit breaker types.					
Shunt trip device	Required for electrically-operated breakers.					
Operations counter	OPT	OPT	OPT	OPT	OPT	OPT
Trip on power up (TOPU) device	N/A	N/A	N/A	N/A	N/A	N/A
Rack-in/rack-out position indicator	STD	STD	STD	STD	STD	STD

NOTES:

/1 Voltage ratings apply to main current-carrying parts. Operating attachments shall be of voltages specified in 6.2 in MIL-DTL-17587.

/2 Ampere values apply to circuit breakers with overcurrent coils or sensors of maximum current rating within the circuit breaker type.

/3 Three-pole design for magnetic effect, two outer poles connected to same circuit polarity surrounding center pole of opposite circuit polarity.

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TABLE II. Available continuous current ratings and pickup current settings for overcurrent coils and sensors.

Type	Coil or sensor rating /1 (amperes)	Long-time pickup setting (amperes)				Short-time pickup setting (amperes)						Maximum instantaneous trip /2 (amperes)
ACB-901R	400	---	---	---	---	---	800	---	---	---	---	2400
	480		---	---	---	---	960	---	---	---	---	2400
	640	960	---	---	---	---	1280	---	---	---	---	3200
	900	1320	---	---	---	---	---	---	2100	---	5400	10000
	1200	---	---	---	---	---	---	---	---	---	6000	10000
ACB-2601R ACB-2603R ACB-2801R	1600	---	---	---	---	---	3200	---	---	---	---	8000
	2600	---	---	---	---	4000	---	---	---	---	---	14000
	2600	---	---	---	---	---	---	---	---	10000	12000	22000
	2800	---	---	---	---	4000	---	---	---	---	---	14000
ACB-4001R /3	400	800	1200	1600	2000	800	960	1000	1200	1600	2000	4000
	800	1600	2400	3200	4000	1600	2000	2400	3200	4000	5000	8000
	1200	2400	3600	4800	6000	2400	3000	3600	4800	6000	----	12000
	2000	4000	6000	8000	10000	4000	5000	6000	7000	8000	10000	20000
	4000	8000	12000	16000	20000	8000	10000	12000	16000	20000	----	40000
ACB-5003R	800	-- 800 1040 2000	--- 880 1200 ---	720 960 1600 ---	--- --- --- ---	800 2800 --- ---	960 3200 --- ---	1200 3600 --- ---	1600 4000 --- ---	2000 5000 --- ---	2400 --- --- ---	12000 --- --- ---
	1200	840 1200 1560 3000	960 1320 1800 ---	1080 1440 2400 ---	--- --- --- ---	1200 4200 --- ---	1440 4800 --- ---	1800 5400 --- ---	2400 6000 --- ---	3000 --- --- ---	3600 --- --- ---	18000 --- --- ---
	2000	1400 2000 2600 5000	1600 2200 3000 --	1800 2400 4000 ---	--- --- --- ---	2000 7000 --- ---	2400 8000 --- ---	3000 9000 --- ---	4000 10000 --- ---	5000 --- --- ---	6000 --- --- ---	30000 --- --- ---
	4000	2800 4000 5200 10000	3200 4400 6000 ---	3600 4800 8000 ---	--- --- --- ---	4000 14000 --- ---	4800 16000 --- ---	6000 18000 --- ---	8000 20000 --- ---	10000 --- --- ---	12000 --- --- ---	50000 --- --- ---

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TABLE II. Available continuous current ratings and pickup current settings for overcurrent coils and sensors - Continued.

Type	Coil or sensor rating /1 (amperes)	Long-time pickup setting (amperes)				Short-time pickup setting (amperes)						Maximum instantaneous trip /2 (amperes)
ACB-5003R	5000	3500	4000	4500	---	5000	6000	7500	10000	12500	15000	50000
		5000	5500	6000	---	17500	20000	22500	25000	---	---	---
		6500	7500	10000	---	---	---	---	---	---	---	---
		12500	---	---	---	---	---	---	---	---	---	---

NOTES:

/1 Continuous current rating.

/2 The instantaneous trip device shall be set at the place of manufacture from five times the coil or sensor rating to the maximum pickup setting specified for mechanical coils and 20 times sensor rating for solid-state sensors with a default at maximum listed rating.

/3 The maximum allowable current through the center non-automatic pole with 1400-amp DC in the two outer poles. (applicable to 4001R)

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TABLE III. Electrical endurance and temperature rise test requirements.

Electrical endurance test:

Type	Total number of close-open cycles	Amperes on circuit breaker close	Amperes on circuit breaker open	DC time constant
ACB-901R	4,000	900	900	0.02 to 0.06
ACB-2601R	1,000	2,600	2,600	
ACB-2603R	1,000	2,600	2,600	
ACB-2801R	1,000	2,800	2,800	
ACB-4001R	4,000	4,000	4,000	
ACB-5003R	1,000	5,000	5,000	0.02 minimum

Temperature rise test:

Type	Continuous current rating	Cable quantity per terminal /1	Cable size	Bus bar quantity per phase	Bus bar size (inches) /2
ACB-901R /3	900	3	300 MCM	1	2.5 x ¼
ACB-2601R	2,600	7 /4	400 MCM	3	3.0 x ¼
ACB-2603R					
ACB-2801R					
ACB-4001R	4,000	11 /4	400 MCM	4	6.0 x ¼
ACB-5003R /5	5,000	13	400 MCM	4	7.0 x ¼

NOTES:

/1 Information regarding cable is contained in MIL-DTL-915. If 3-phase cable is used for this test, the cable jacket shall be stripped for a minimum of 2 feet.

/2 Information regarding bus bars is contained in MIL-DTL-16036.

/3 For an ACB-901R with a 1200-amp coil rating, the acceptance criteria shall be 10 °C greater than those specified in Table I of MIL-DTL-17587(SH).

/4 Connect with bus adapter of minimum length possible.

/5 For the ACB-5003R, the following temperature rise criteria exceptions to Table I of MIL-DTL-17587(SH) apply:

a. Pre-interruption at rated current: Maximum temperature rise above enclosure external ambient shall be 85 °C for terminals, contacts, and conducting mechanical joints. Maximum temperature with 50 °C enclosure external ambient shall be 135 °C for terminals, contacts, and conducting mechanical joints.

b. Pre-interruption at 150% rated current and 5-minute duration: Maximum temperature rise above enclosure external ambient shall be 90 °C for terminals and conducting mechanical joints. Maximum temperature with 50 °C enclosure external ambient shall be 140 °C for terminals and conducting mechanical joints.

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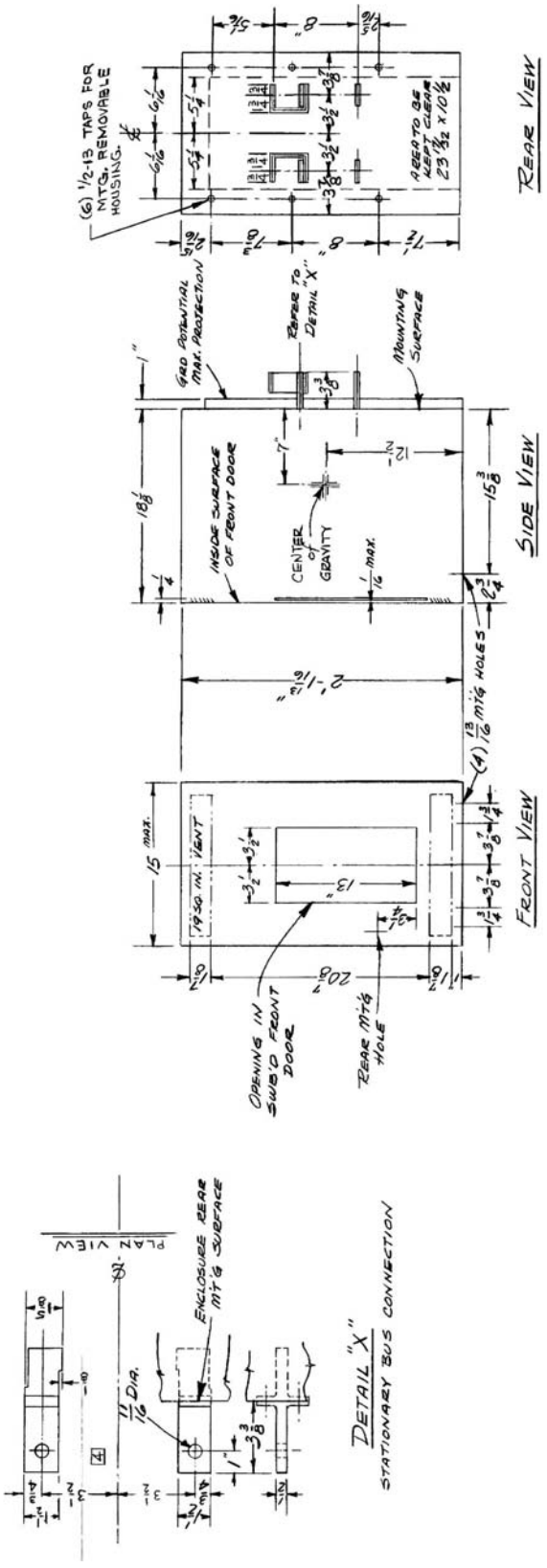


FIGURE 1. Outline and mounting dimensions for circuit breaker type ACB-901R.

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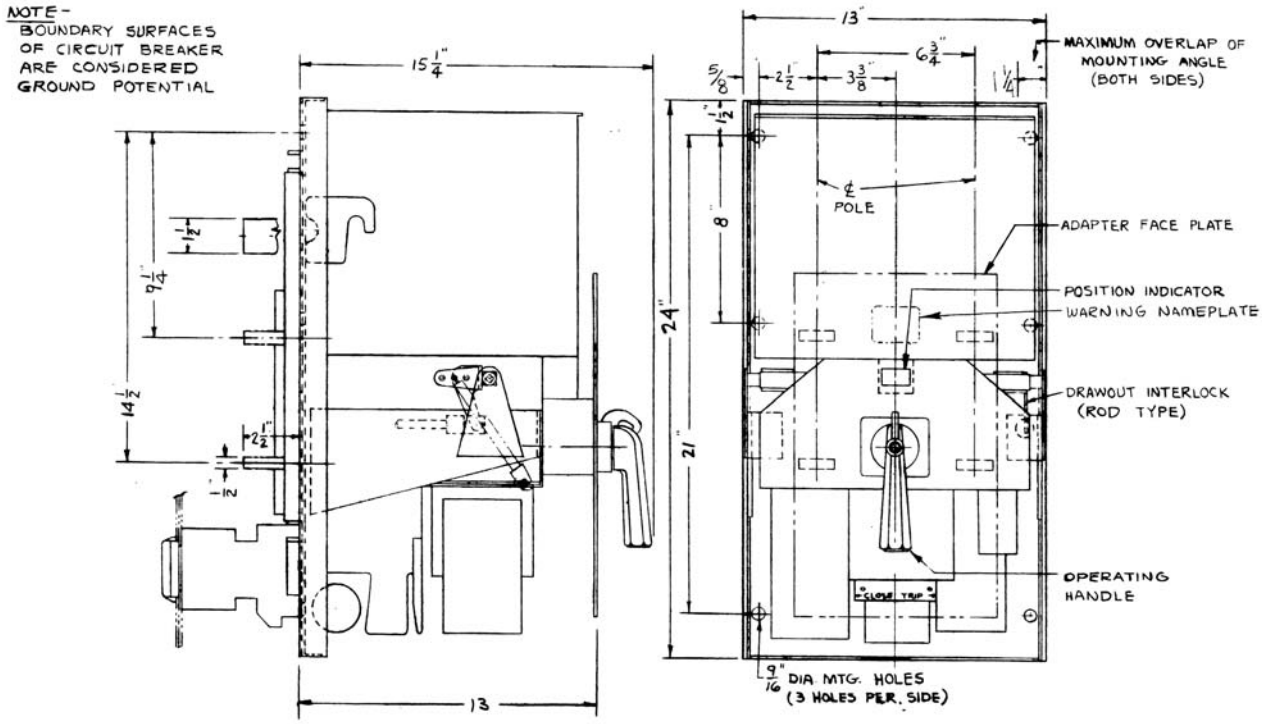


FIGURE 2. Outline view of moving component for circuit breaker type ACB-901R.

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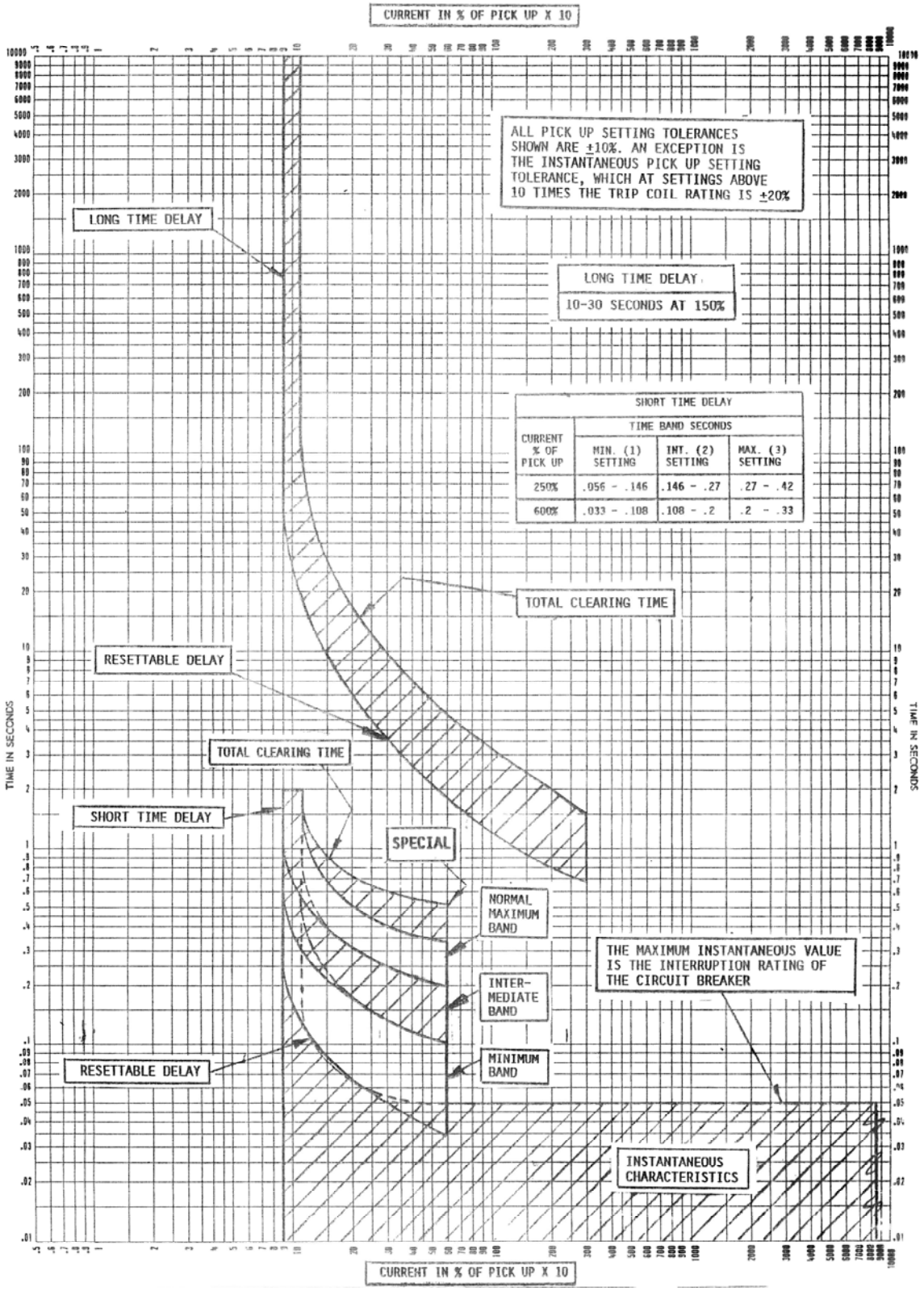


FIGURE 3. Time-current characteristic curve for circuit breaker type ACB-901R.

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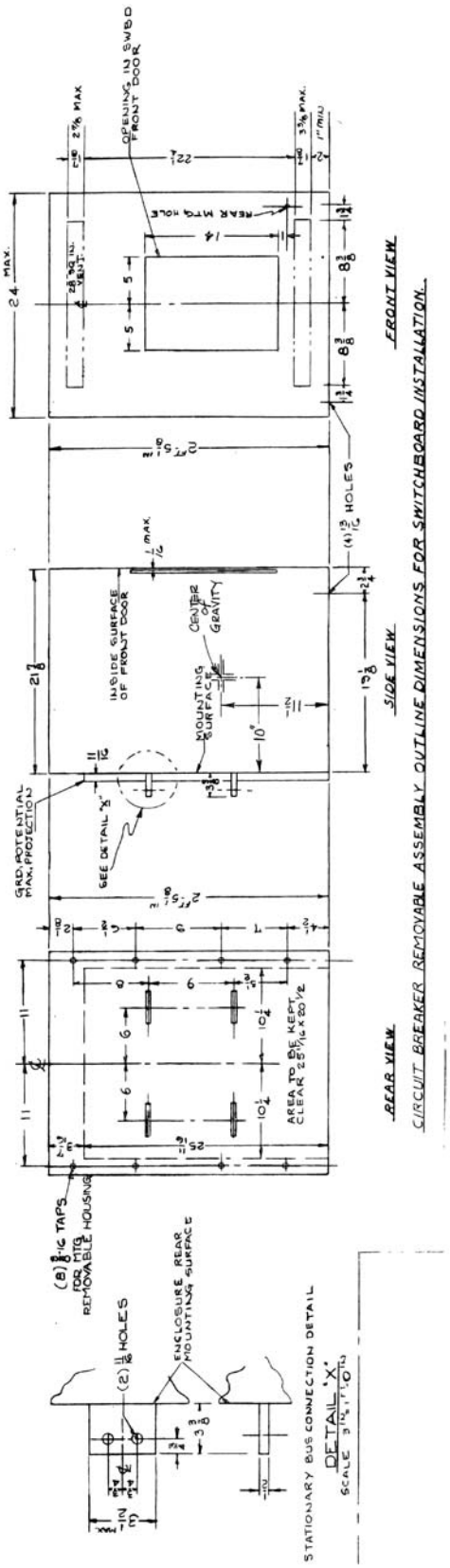


FIGURE 4. Removable assembly for circuit breaker types ACB-2601R and ACB-2801R.

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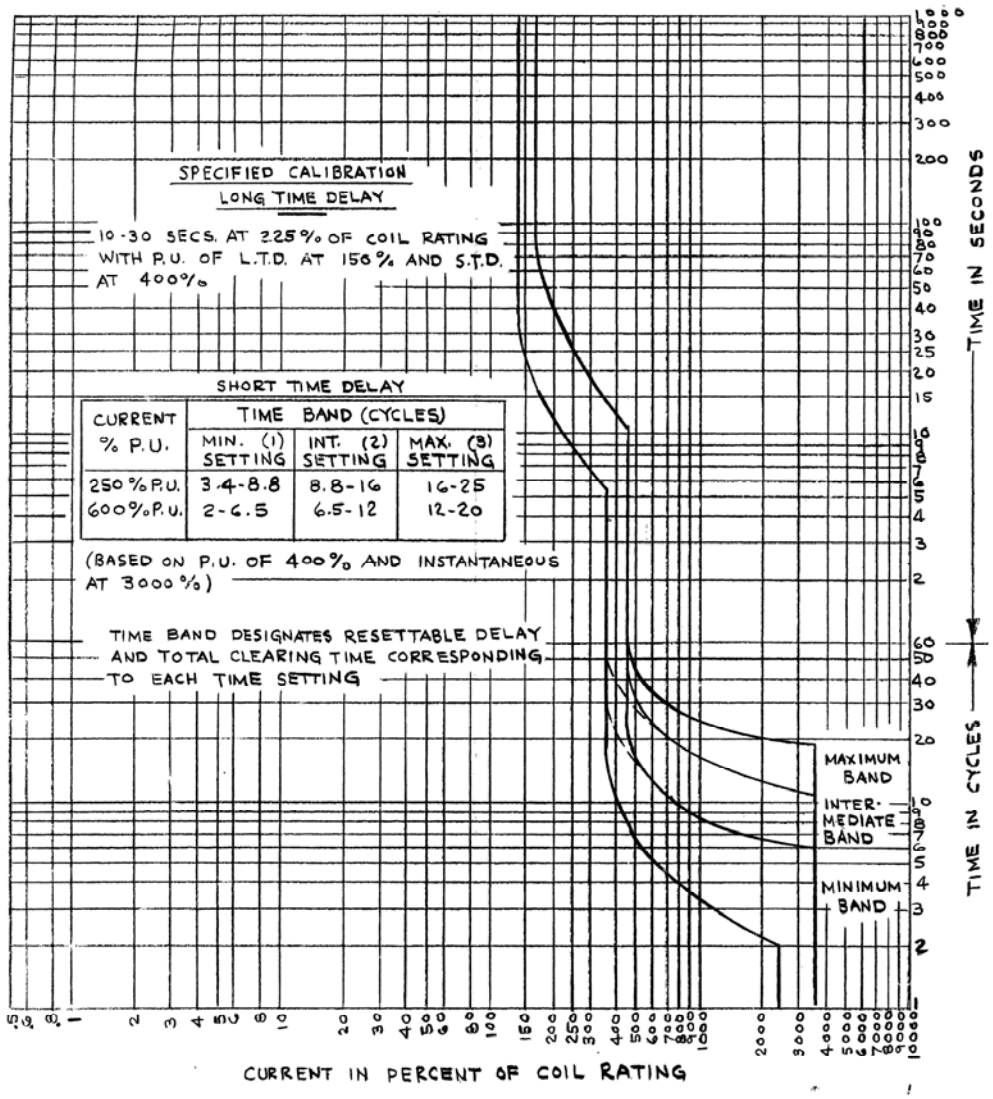


FIGURE 5. Time-current characteristic curve for circuit breaker types ACB-2601R and ACB-2801R.

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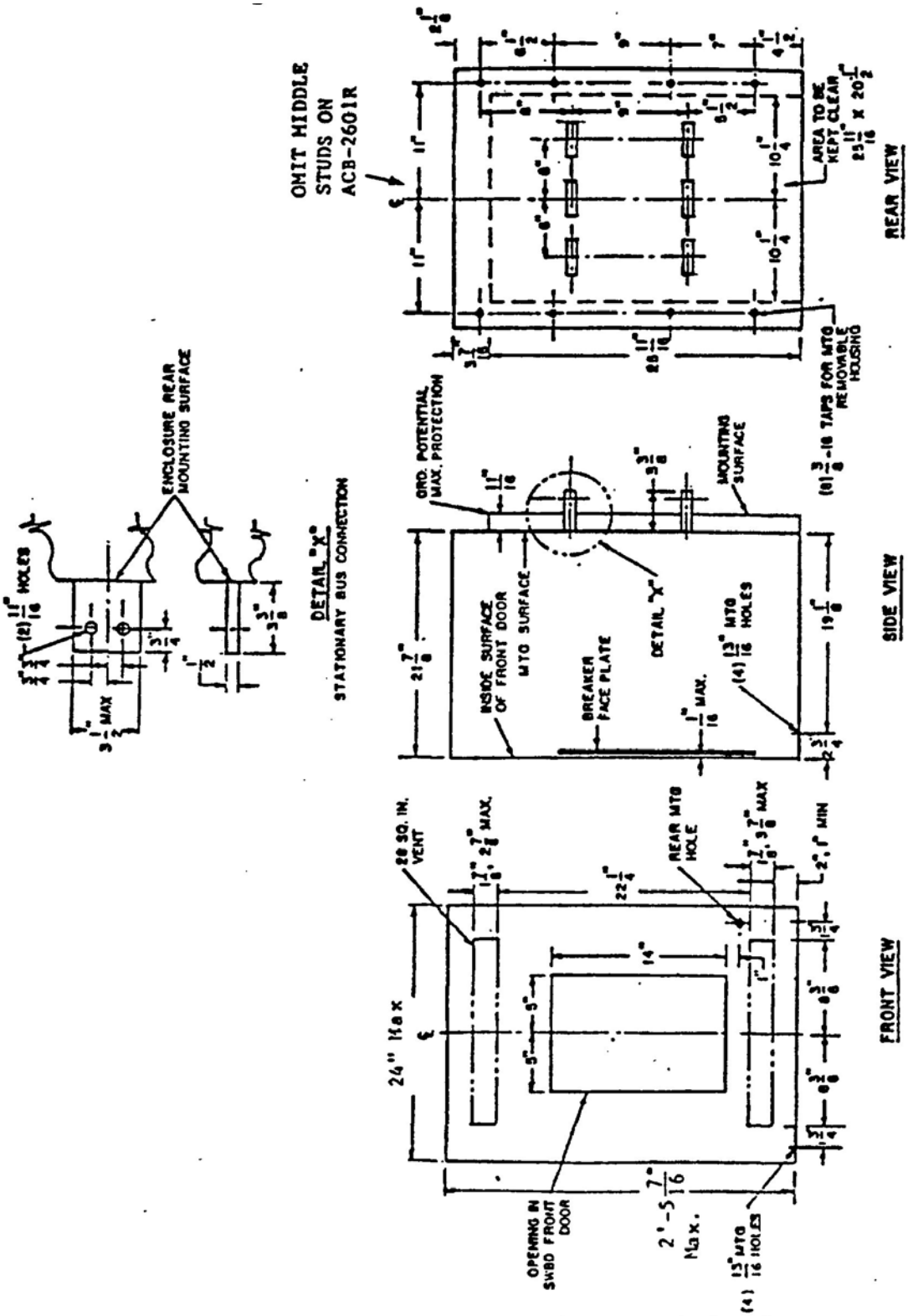


FIGURE 6. Outline and mounting dimensions for circuit breaker type ACB-2603R.

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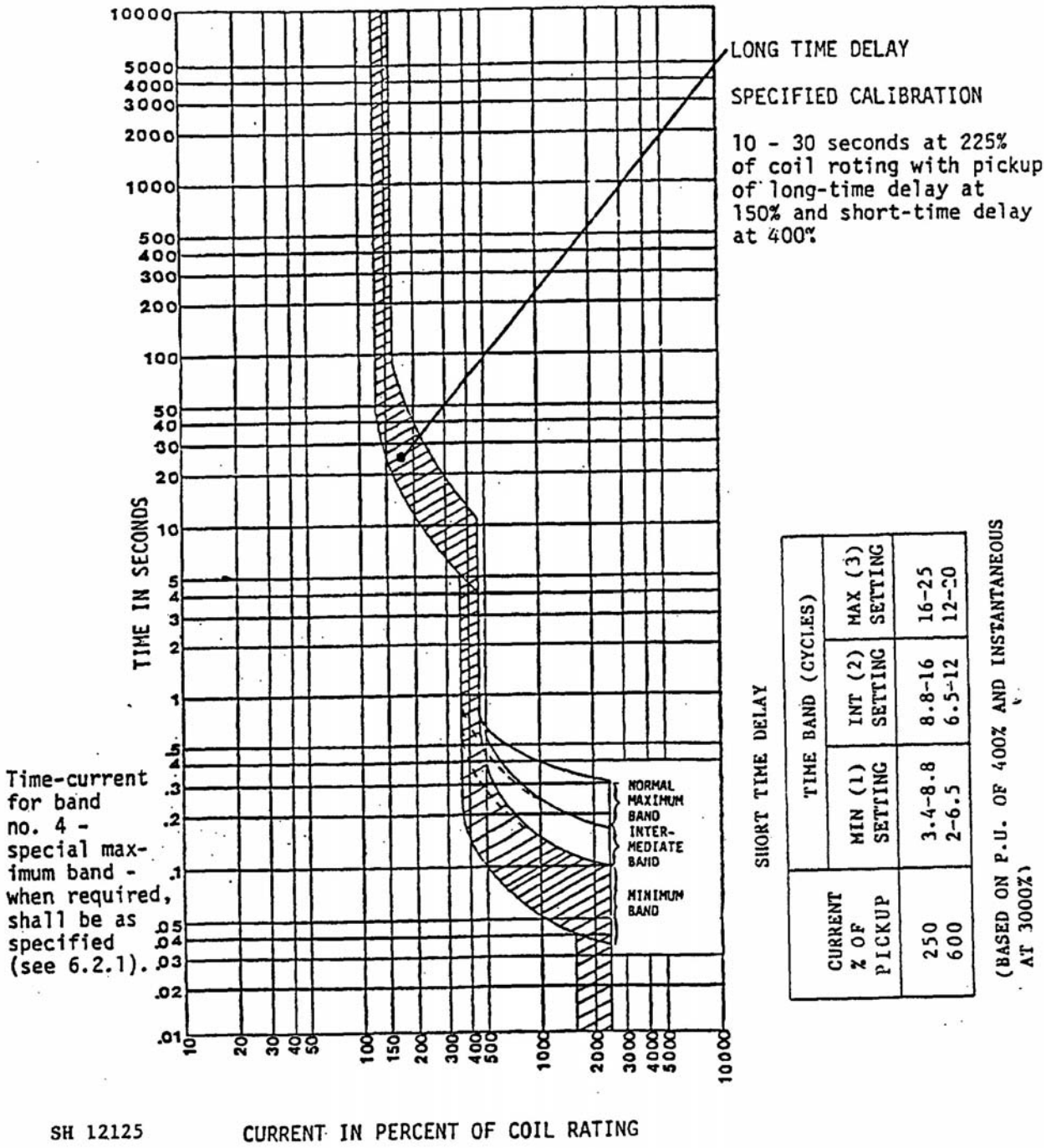


FIGURE 7. Time-current characteristic curve for circuit breaker type ACB-2603R.

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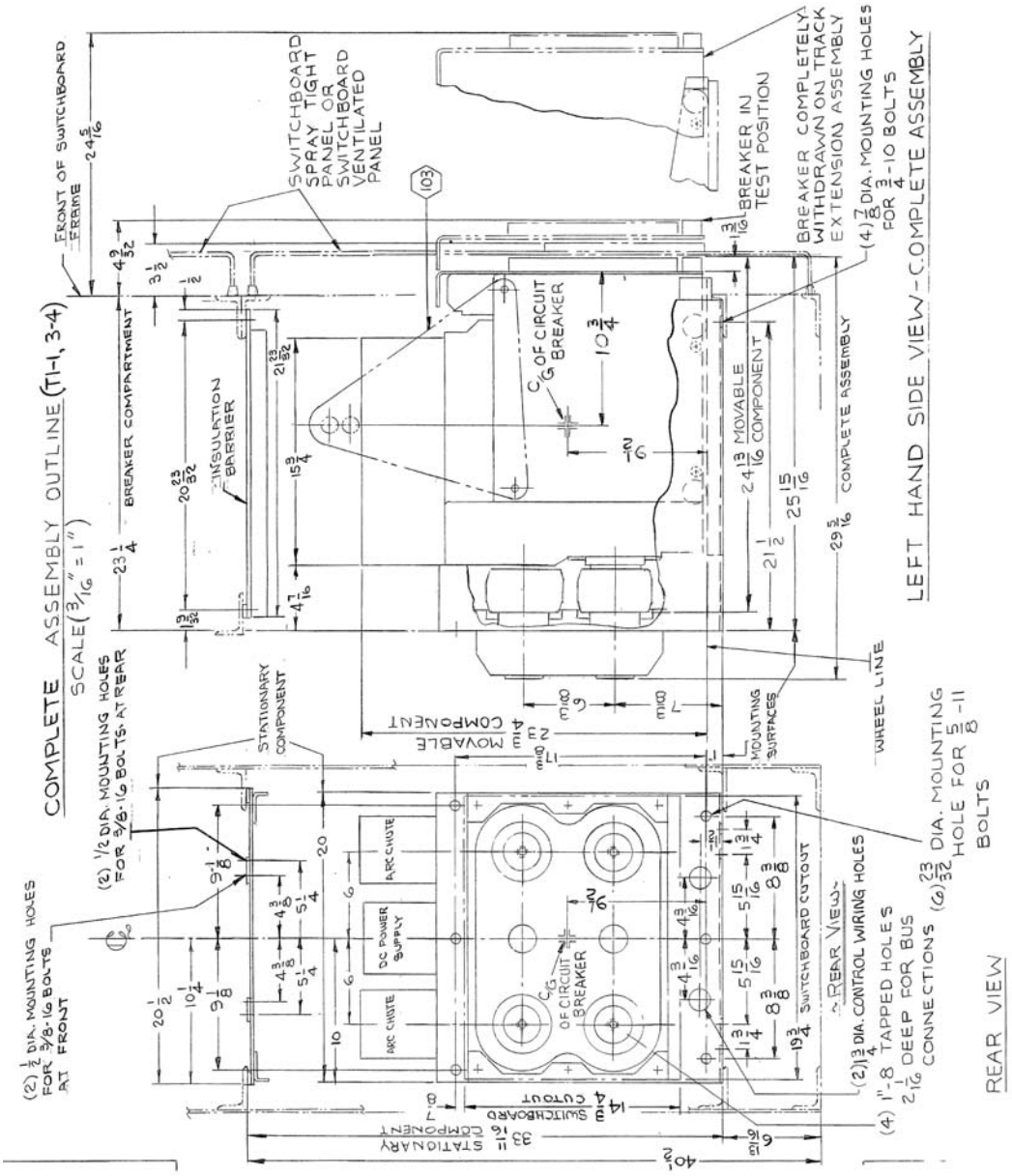


FIGURE 8. Outline and mounting dimensions for circuit breaker type ACB-4001R.

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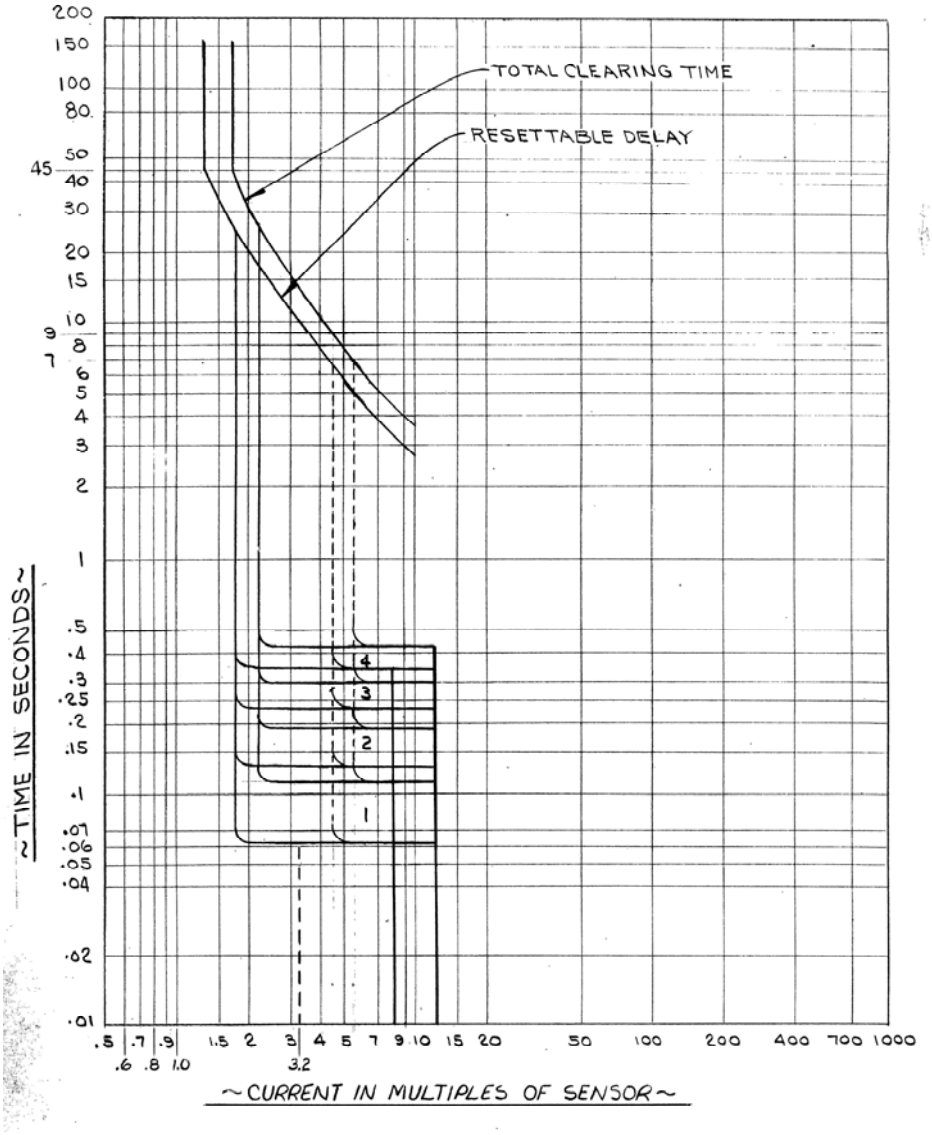
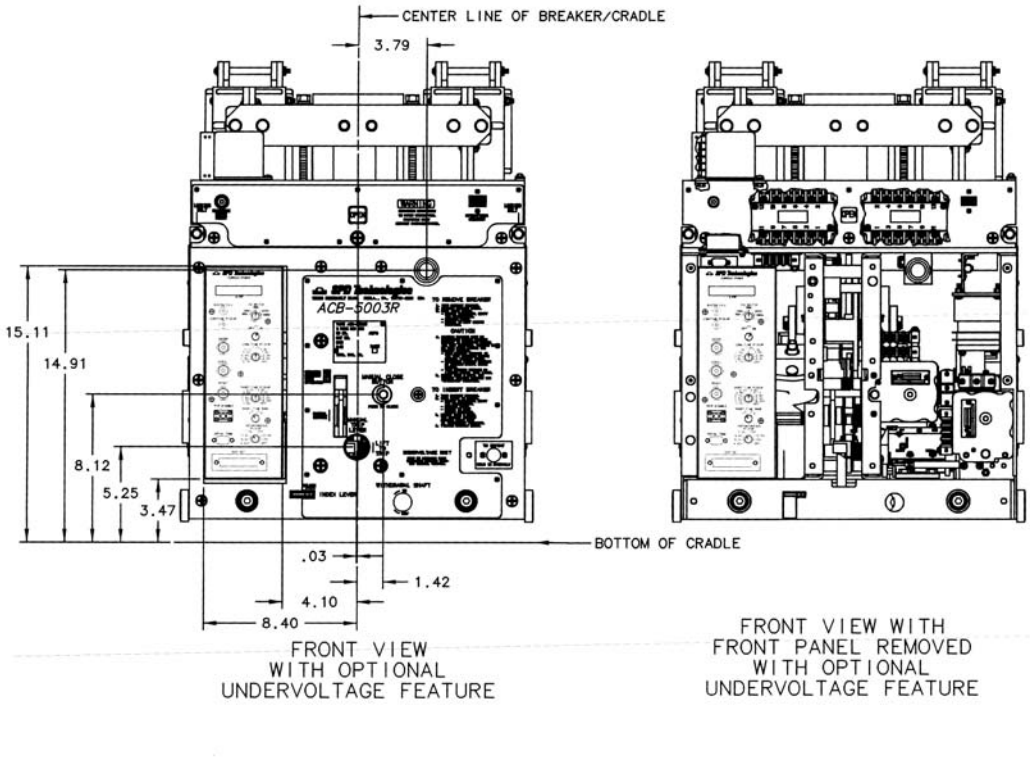


FIGURE 9. Time-current characteristic curve for circuit breaker type ACB-4001R.

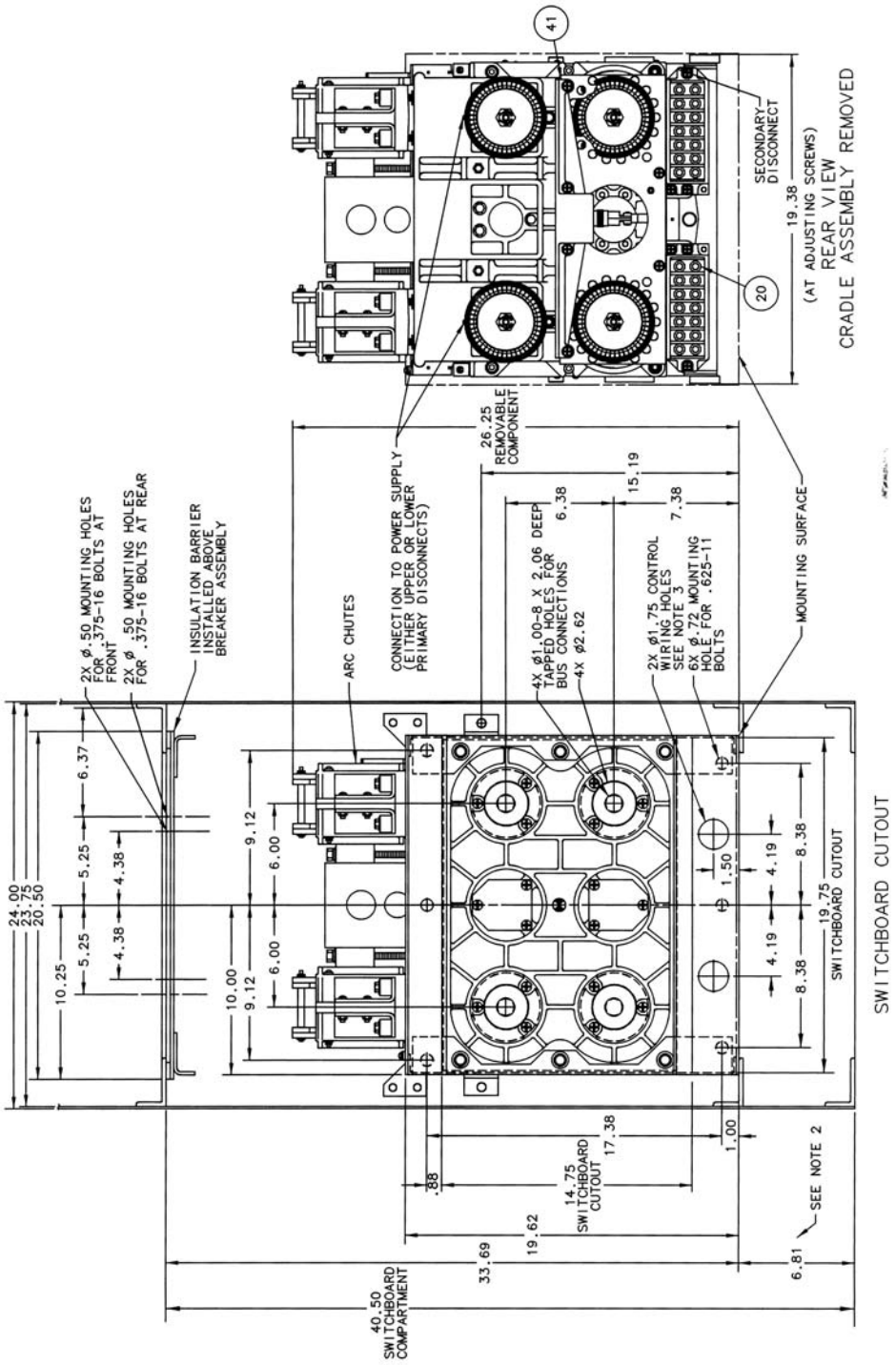
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NOTE: Front view.

FIGURE 10. Outline and mounting dimensions for circuit breaker type ACB-5003R.

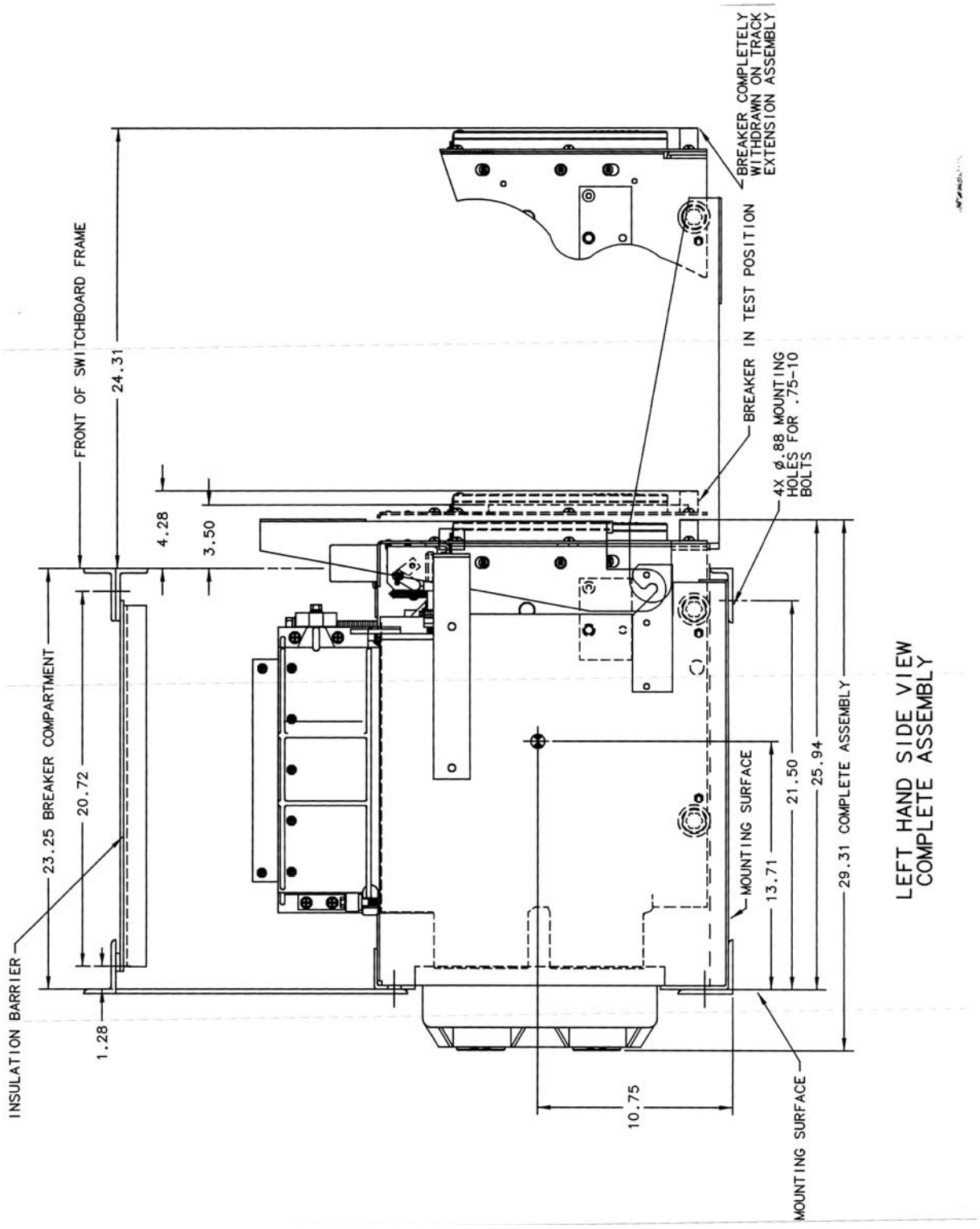
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NOTE: Rear view.

FIGURE 10. Outline and mounting dimensions for circuit breaker type ACB-5003R - Continued.

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NOTE: Left-hand side view.

FIGURE 10. Outline and mounting dimensions for circuit breaker type ACB-5003R - Continued.

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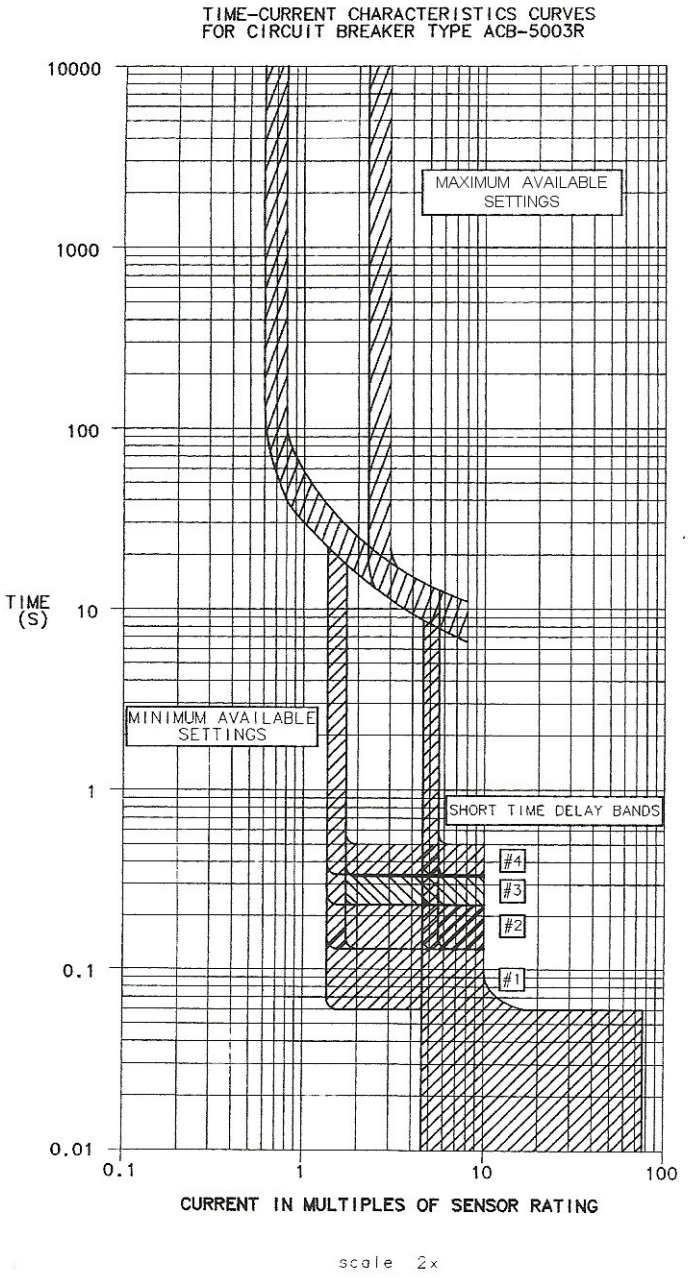


FIGURE 11. Time-current characteristic curve for circuit breaker type ACB-5003R.

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